

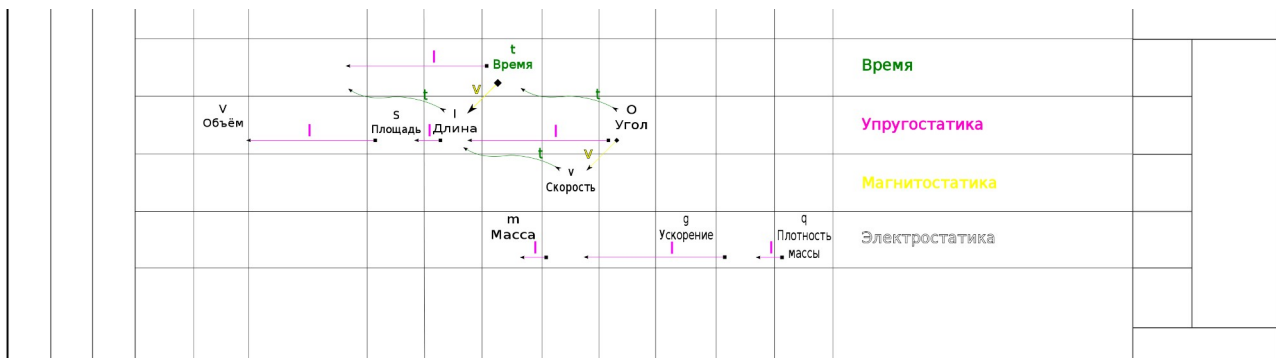
ALGORITHM FOR AUTOMATED SOLUTION OF ELEMENTARY PHYSICS PROBLEMS (7TH GRADE)

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Based on the 7th-grade basic physics curriculum, a graphical representation of physical formulas is formed. A table of physical quantities is constructed from chains of these formulas. The proposed table allows replacing the standard general list of formulas with a unified concept of using a planar graph. The table contains all the necessary connections between physical quantities, their symmetric relationships, as well as their tensor representation.

The table facilitates the memorization of physical laws and allows for the convenient application of various algebraic tools (real and imaginary numbers, orthogonal functions, matrices, quaternions, tensors, differential forms, systems of differential equations, and the relationships between them) for solving specific physical problems.



The structure of the table includes the following blocks: volume, area, length, time, angle, elastostatics, speed, mass, acceleration, density, and electrostatics.

This structure serves as the basis for representing data in a computer program designed to search for typical solutions to elementary physics problems (7th grade). In addition to physical formulas, these problems utilize several mathematical models, in particular, the volume of a solid parallelepiped (which is equal to the product of its 3 geometric dimensions), as well as the area of a circle, the surface area of a sphere, and the volume of a sphere.

A wave algorithm for finding the shortest path between the cells of the table of physical quantities is proposed as a formula computation algorithm. The result of this operation is a chain of transformations of physical quantities during the transition from one to another. As a rule, these transformations represent parallel translation in space-time, spatial rotation, and boosts (Lorentz rotations). The "star" operator (Hodge star operator) should also be added here.

Each link in such a transformation chain is used as a basic rule for an elementary physical formula that includes 2 physical quantities in explicit form and one as a coefficient or

constant. Moreover, each such formula is included in the working list of physical formulas 3 times, depending on the required quantity (1 out of 3).

Through sequential substitution (symbolic computation) and simplification, it is possible to obtain the final formula required to solve a physics problem, which is successfully implemented in this script-program. Thus, this report describes an algorithm for deriving a physical formula based on the table of physical quantities.

A development of this system could be the tight integration of such an algorithm into CAD (computer-aided design) systems. In the future, it may even be possible to actively input problem conditions via a 3D interface for planning virtual experiments and setting initial physical conditions for the desired system.

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